

WORTHINGTON, MN, USA

WMO: 726587

Lat: 43.645N

Lon: 95.580W

Elev: 479

StdP: 95.71

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94927

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.0	-22.0	-29.1	0.3	-20.8	-26.7	0.3	-19.6	16.1	-9.5	14.7	-8.2	5.0	280	0.671

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	31.5	22.7	29.9	21.9	28.1	20.8	24.7	29.2	23.7	28.0	22.6	26.7	6.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.0	18.9	27.5	22.3	18.0	26.9	21.2	16.8	25.8	78.0	29.1	73.2	28.0	68.8	26.5	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.8	(b) 12.1	(c) 11.0	(e) -27.2	(f) 34.5	(g) 2.6	(h) 1.7	(i) -29.1	(j) 35.7	(k) -30.6	(l) 36.7	(m) -32.0	(n) 37.7	(o) -33.9	(p) 38.9		
(4) 13.8	(5) 12.1	(6) 11.0	(7) -27.6	(8) 26.4	(9) 2.5	(10) 1.5	(11) -29.5	(12) 27.4	(13) -30.9	(14) 28.3	(15) -32.4	(16) 29.1	(17) -34.2	(18) 30.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.4	-9.0	-6.7	0.1	7.4	13.8	19.8	21.8	20.2	16.5	9.2	1.0	-6.2		
	DBStd	12.15	7.25	7.07	6.95	5.55	5.01	3.92	3.23	3.12	4.72	5.46	6.24	6.55		
	HDD10.0	2362	588	467	314	110	19	0	0	0	6	82	275	502		
	HDD18.3	4309	847	700	566	328	158	29	8	17	90	288	519	760		
	CDD10.0	1417	0	0	7	33	138	294	367	317	200	56	6	0		
	CDD18.3	321	0	0	0	2	18	73	117	75	34	3	0	0		
	CDH23.3	2782	0	0	1	31	208	664	1005	542	287	44	1	0		
	CDH26.7	779	0	0	0	7	67	212	293	122	69	9	0	0		
(14) Wind	WSAvg	5.4	6.0	6.1	6.0	6.2	6.0	5.0	4.1	3.8	4.5	5.4	5.8	5.8		
(15) Precipitation	PrecAvg	726	14	18	34	70	91	120	86	84	78	50	27	19		
	PrecMax	1107	37	38	87	186	166	284	237	243	233	132	91	59		
	PrecMin	466	0	2	0	9	26	47	9	16	11	6	0	3		
	PrecStd	151	11	10	24	40	36	49	56	58	65	38	27	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	12.7	21.3	27.3	32.4	33.4	33.0	32.1	31.2	27.7	20.2	11.2		
		MCWB	3.5	8.7	13.0	16.1	19.4	22.2	24.9	24.7	21.3	17.2	12.1	8.0		
	2%	DB	3.8	7.8	17.1	22.6	27.9	31.2	31.4	29.1	28.7	23.7	16.1	7.3		
		MCWB	1.3	5.2	11.4	13.2	17.5	21.2	24.2	22.8	20.9	15.0	10.4	4.1		
	5%	DB	2.3	4.6	12.7	19.0	25.9	28.9	29.6	27.8	27.0	20.9	12.7	3.8		
		MCWB	0.6	2.4	8.4	11.3	16.5	20.1	23.2	21.7	19.1	14.1	8.1	1.2		
	10%	DB	0.9	2.5	9.1	16.4	22.5	27.3	27.9	26.4	24.6	17.7	9.8	2.3		
		MCWB	-0.5	0.8	6.2	9.8	14.8	18.9	21.9	20.6	18.3	12.1	6.2	0.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	9.5	14.9	17.0	21.3	24.6	26.9	25.6	23.5	19.6	13.9	8.6		
		MCDB	7.3	12.6	18.3	25.4	28.8	30.2	30.6	29.9	28.1	24.6	18.8	10.0		
	2%	WB	1.8	5.3	11.7	14.5	19.5	23.0	25.3	24.1	21.9	16.9	10.9	3.9		
		MCDB	3.5	7.9	16.2	20.9	25.8	28.4	29.8	28.3	26.5	21.1	14.6	7.0		
	5%	WB	0.5	2.8	9.1	12.5	17.6	21.5	24.2	23.0	20.6	14.8	8.6	1.7		
		MCDB	2.0	4.7	12.9	17.5	23.6	27.2	28.5	26.7	25.3	19.5	12.5	3.6		
	10%	WB	-0.7	0.7	6.3	10.7	15.8	20.2	23.2	21.7	19.3	12.7	6.2	0.3		
		MCDB	0.4	2.2	9.8	15.3	20.6	25.7	27.2	25.4	23.4	17.4	9.7	2.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	8.9	9.0	11.0	11.0	10.5	10.5	10.3	11.6	11.0	9.3	8.1		
		MCDBR	9.7	10.2	13.9	16.0	15.3	12.9	11.2	11.3	13.5	15.1	13.5	10.1		
		MCWBR	8.0	8.1	9.2	9.0	7.4	6.1	6.2	6.7	7.2	8.4	9.0	7.9		
		MCDBR	9.2	10.0	13.4	14.1	12.8	11.4	10.3	10.3	11.3	12.9	12.8	9.5		
	5% WB	MCWBR	7.7	8.1	9.3	8.8	7.1	6.3	6.4	6.8	6.8	8.2	8.9	7.6		
(40) Clear-Sky Solar Irradiance	taub		0.262	0.278	0.309	0.362	0.383	0.398	0.408	0.407	0.361	0.322	0.286	0.267		
	taud		2.446	2.423	2.437	2.320	2.333	2.348	2.308	2.285	2.389	2.486	2.512	2.459		
	Ebn at Noon		885	927	935	902	889	873	859	847	864	862	849	843		
	Edh at Noon		72	88	99	121	124	122	126	125	104	82	68	65		
(44) All-Sky Solar Radiation	RadAvg		1.88	2.98	3.86	4.77	5.52	6.16	6.60	5.46	4.36	2.93	1.99	1.48		
	RadStd		0.19	0.28	0.34	0.43	0.40	0.42	0.25	0.40	0.30	0.37	0.14	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (53 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page